Lecture 8: Schematics of Robots, Modeling a Real Robot

Iñigo Iturrate

Associate Professor

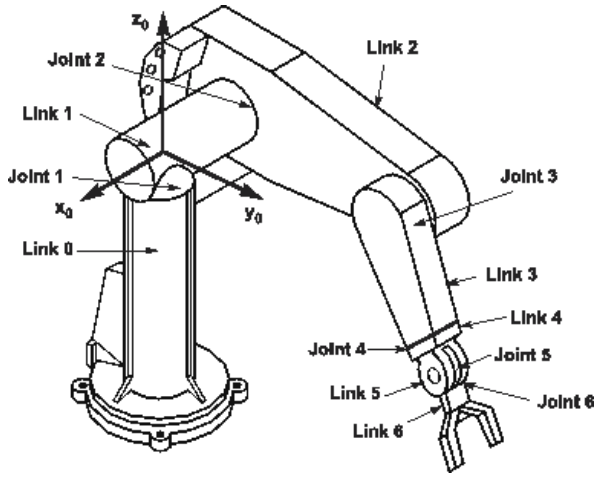
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What are we doing today?

1. Introduction to Robotics & Recap of Linear Algebra and Mathematical Notation
2. Translations & Rotation Matrices
3. Other Representations for Orientation
4. Transformation Matrices
5. Forward Kinematics & DH Parameters
6. Inverse Kinematics
7. Kinematic Simulation
8. **Schematic Representations of Robots, Modeling a Real Robot** ***(Today)***
9. Angular Velocities, Velocity Propagation & Trajectory Generation
10. The Jacobian Matrix, Singularities
11. Analytical & Geometrical Jacobian, Numerical IK, Statics
12. Manipulability, More on the Robotic Systems Toolbox

Recap: DH Parameters

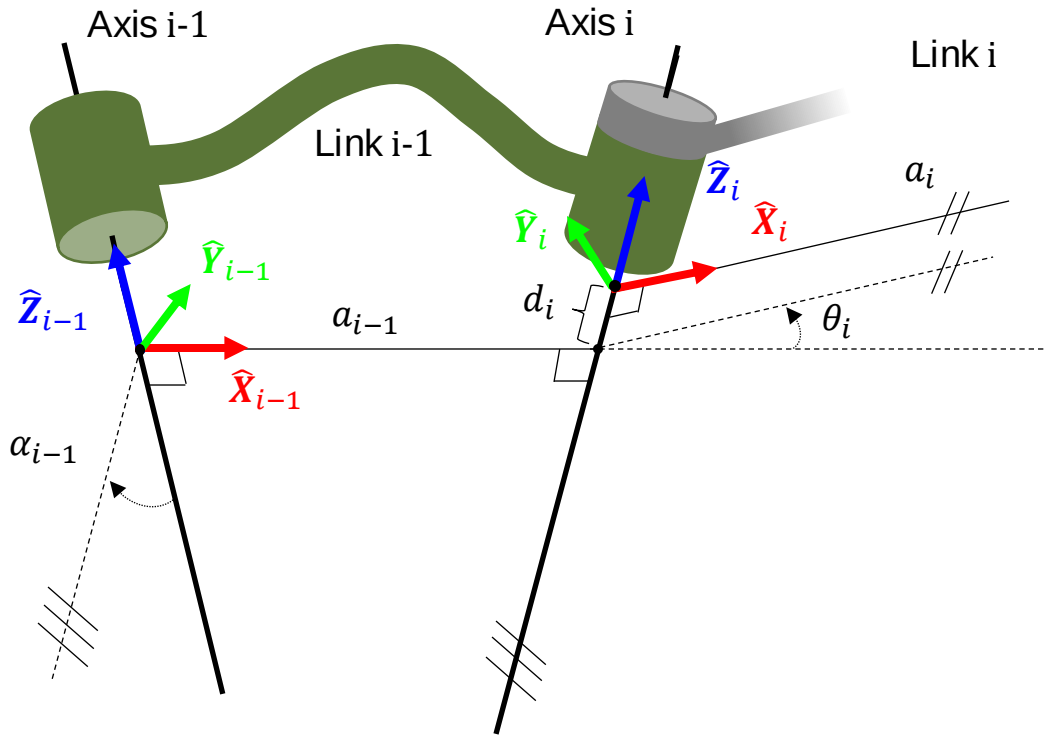
Using **Modified** (Craig) **DH Parameters**:

We can describe a whole robot by **attaching frames to links**:

- **Z-axis** of frame $\{i\}$
  is coincident with joint axis i.
- **X-axis** of frame $\{i\}$
  points along a_i in the direction from joint i to joint $i+1$.

Then, we can **assign four parameters to each link**:

- $\boldsymbol{\alpha_{i-1}}$ – The angle from $\hat{\boldsymbol{Z}}_{i-1}$ to $\hat{\boldsymbol{Z}}_i$ measured about $\hat{\boldsymbol{X}}_{i-1}$
- $\boldsymbol{a_{i-1}}$ – The distance from $\hat{\boldsymbol{Z}}_{i-1}$ to $\hat{\boldsymbol{Z}}_i$ measured along $\hat{\boldsymbol{X}}_{i-1}$
- $\boldsymbol{d_i}$ – The distance from $\hat{\boldsymbol{X}}_{i-1}$ to $\hat{\boldsymbol{X}}_i$ measured along $\hat{\boldsymbol{Z}}_i$
- $\boldsymbol{\theta_i}$ – The angle from $\hat{\boldsymbol{X}}_{i-1}$ to $\hat{\boldsymbol{X}}_i$ measured about $\hat{\boldsymbol{Z}}_i$



Recap: FK from Modified DH Parameters

These variables are always fixed

This is fixed for a revolute joint

This is fixed for a prismatic joint

$$
{}^{i-1}_{i}\boldsymbol{T} = \boldsymbol{R}_X(\alpha_{i-1})\boldsymbol{D}_X(a_{i-1})\boldsymbol{D}_Z(d_i)\boldsymbol{R}_Z(\theta_i)
$$

$$
= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\alpha_{i-1} & -s\alpha_{i-1} & 0 \\ 0 & s\alpha_{i-1} & c\alpha_{i-1} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_{i-1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

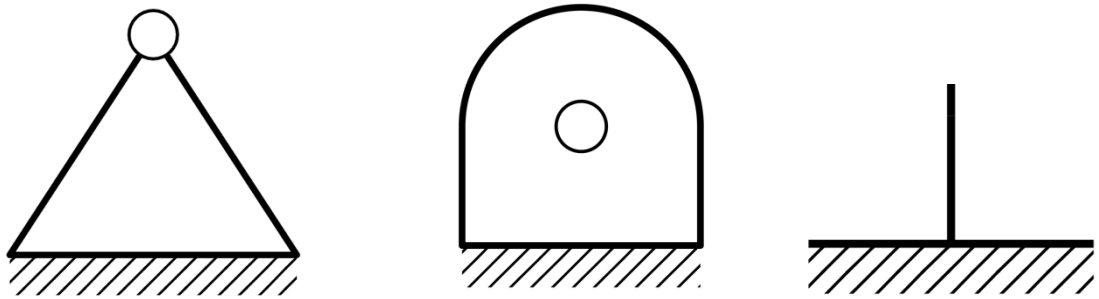
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Recap: MATLAB RigidBodyTree

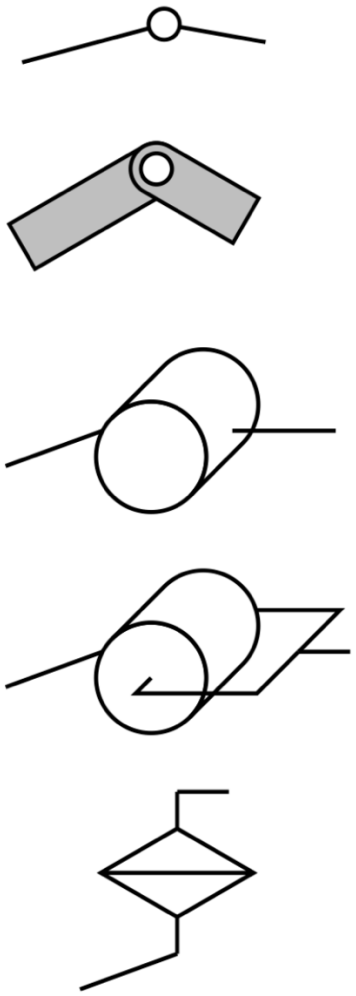
Let us see it in MATLAB

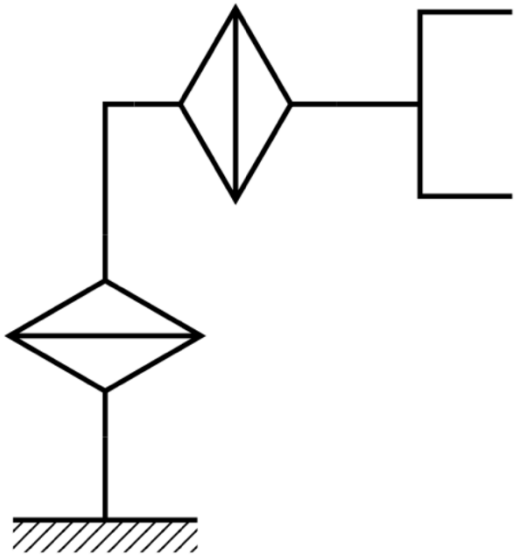
Part I: Schematic Representations of Robot Kinematics

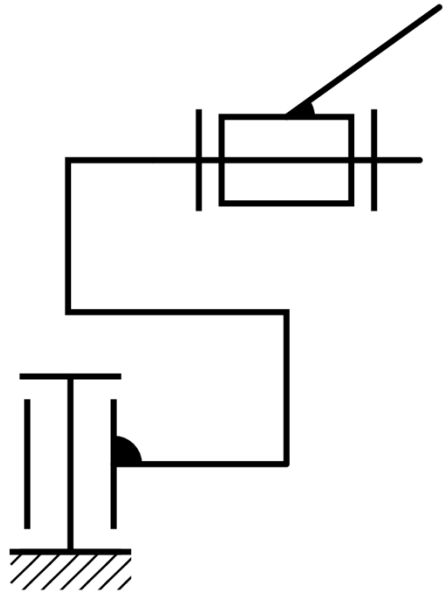
Robot Base



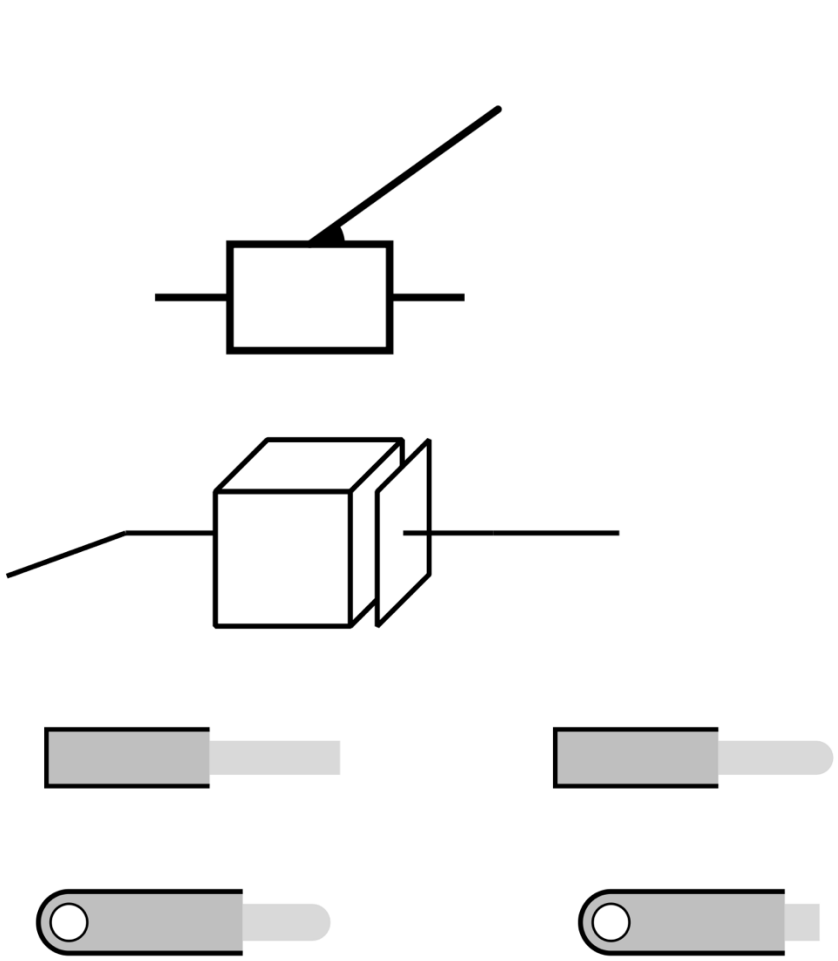
Revolute Joints

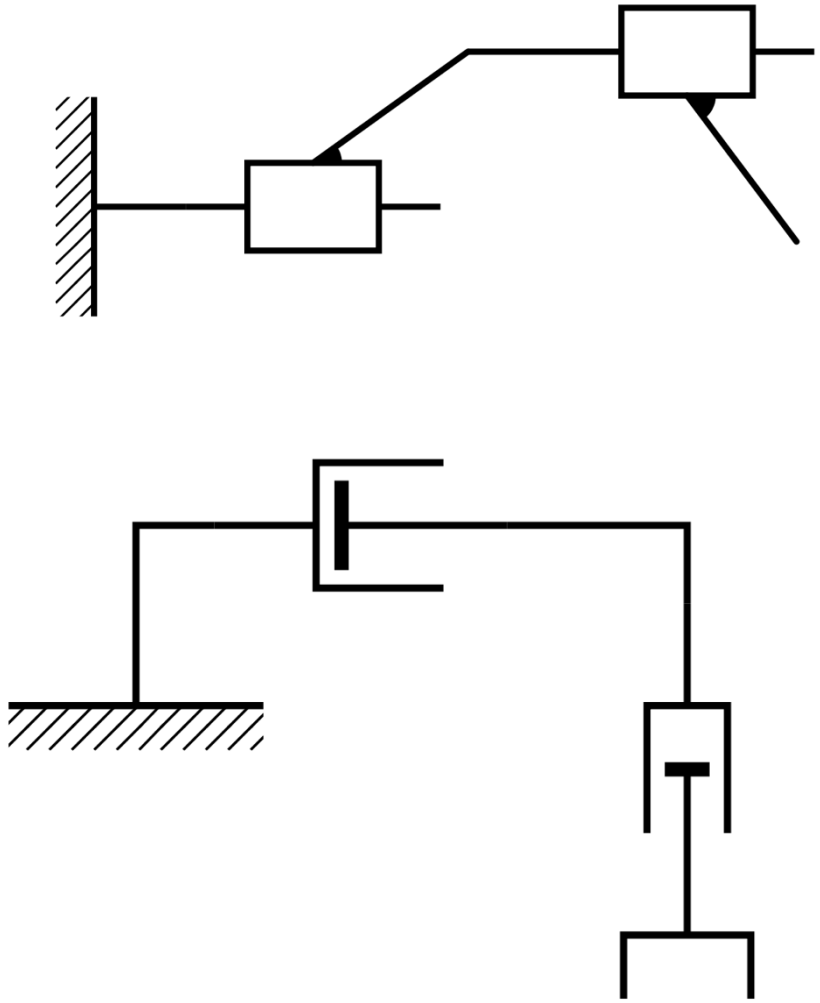






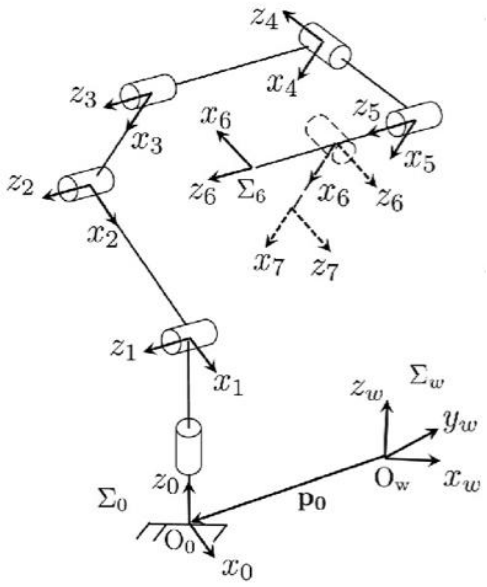
Prismatic Joints

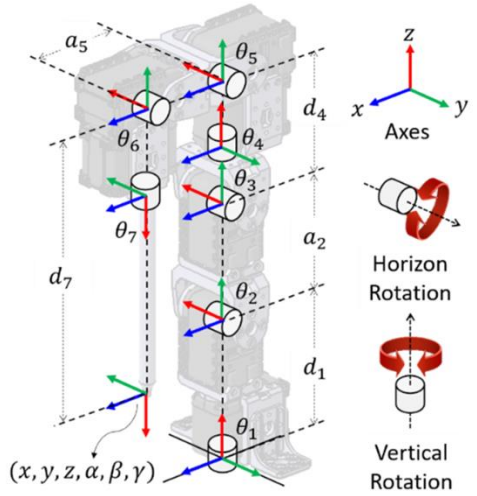


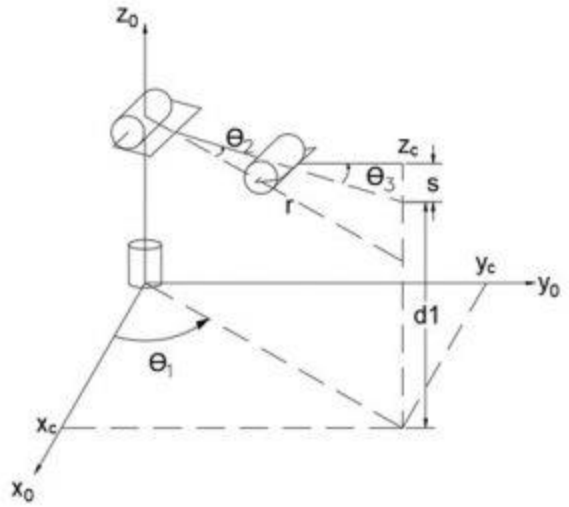


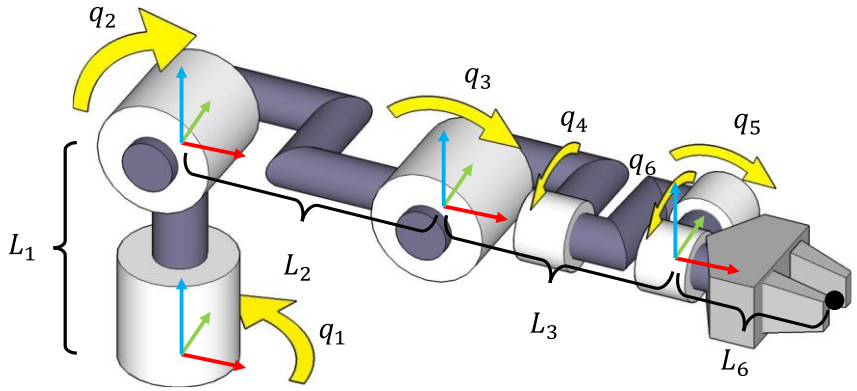
Examples (I)

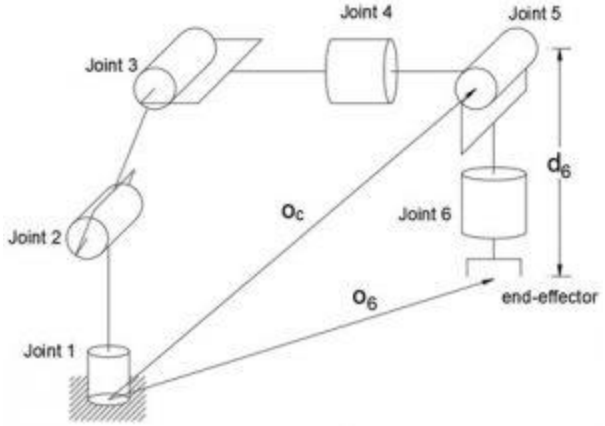












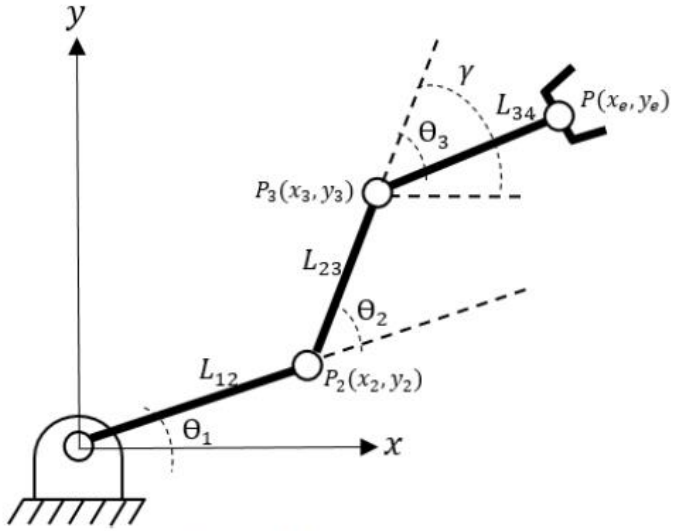
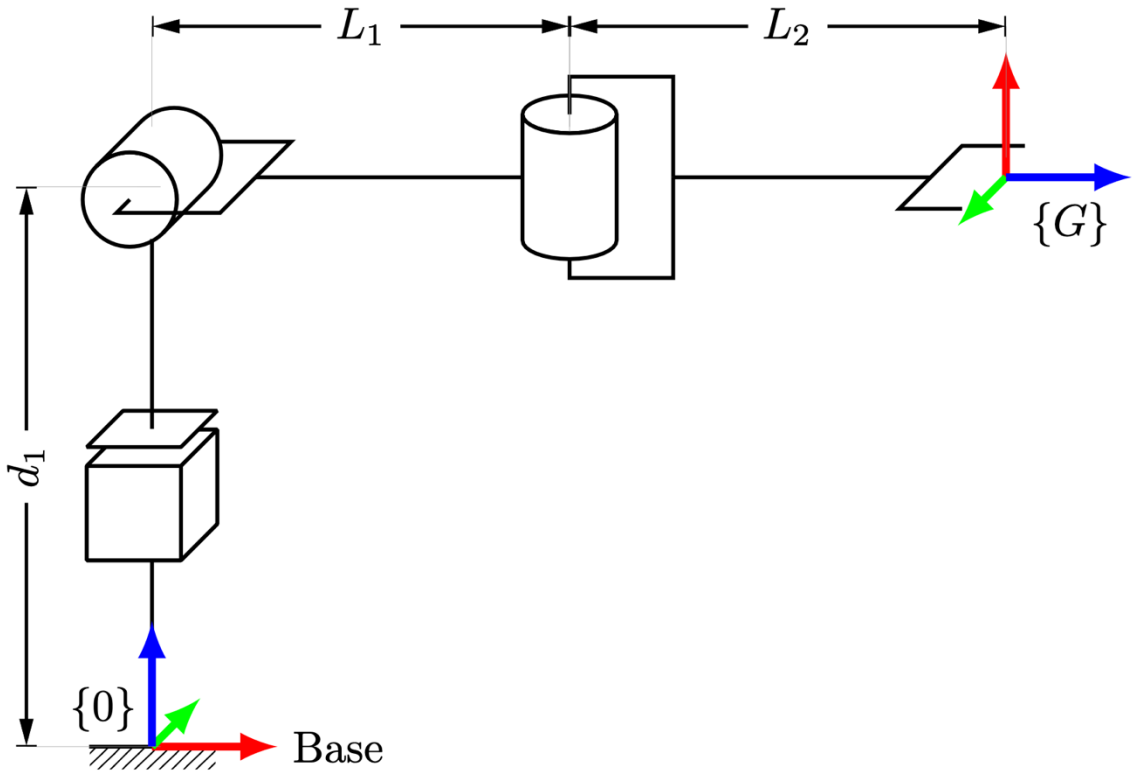
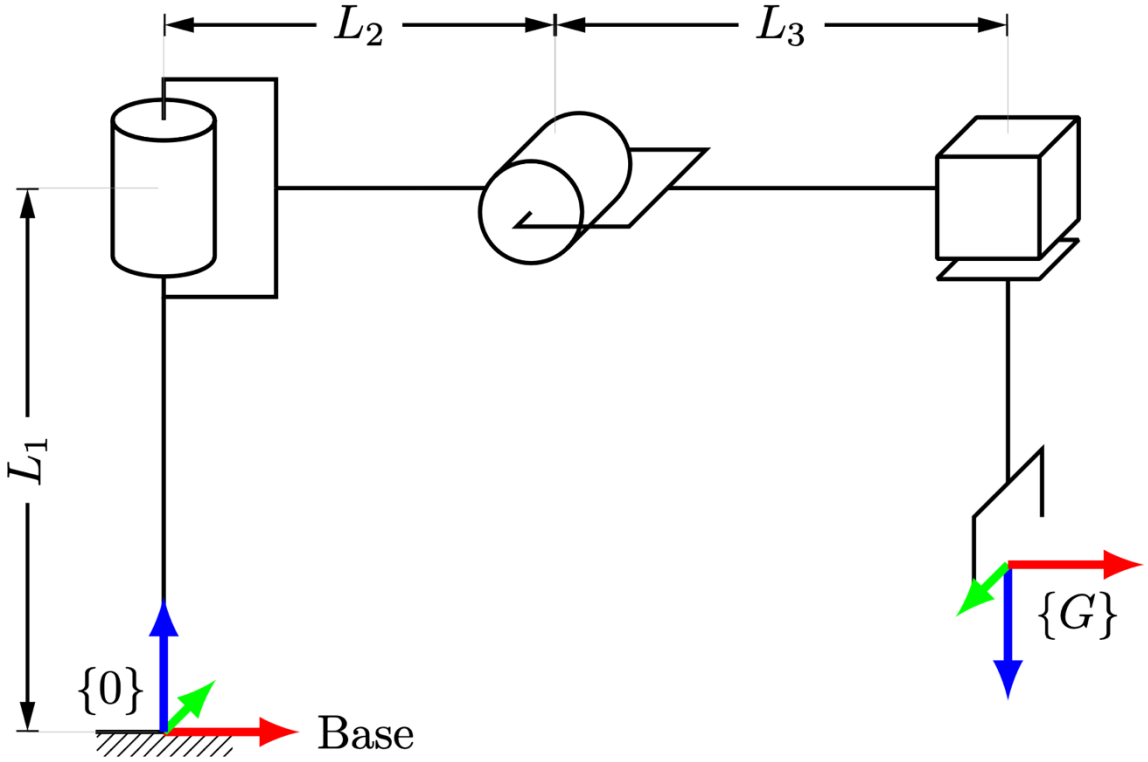


Fig. 1: 3R Planar Manipulator

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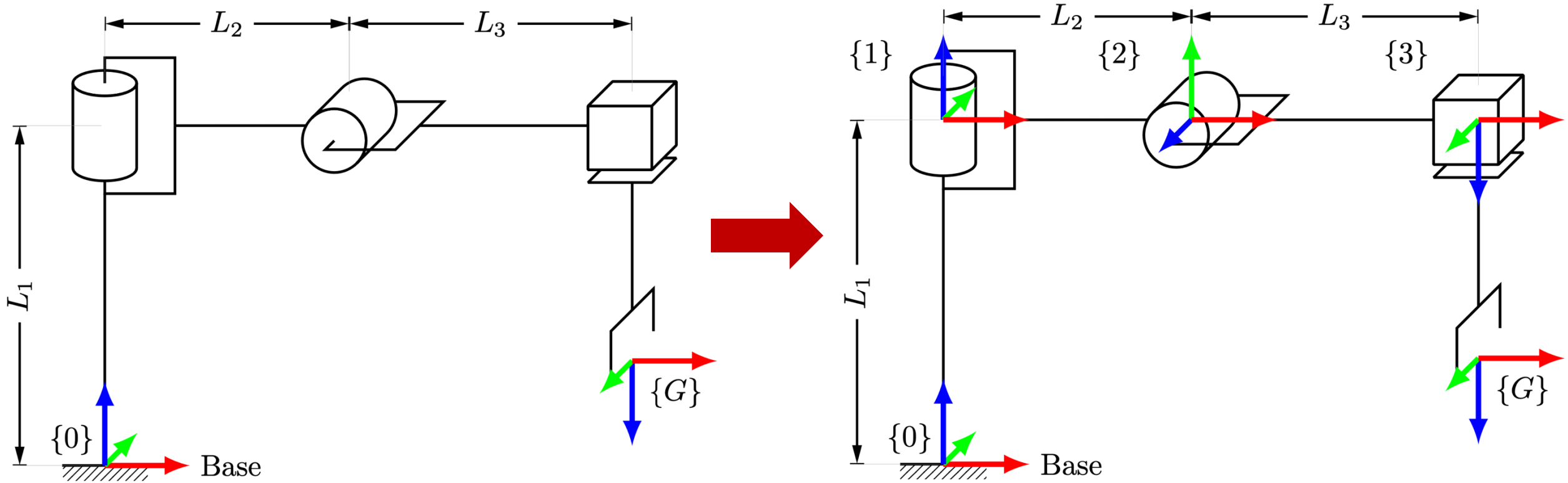
Examples (II)



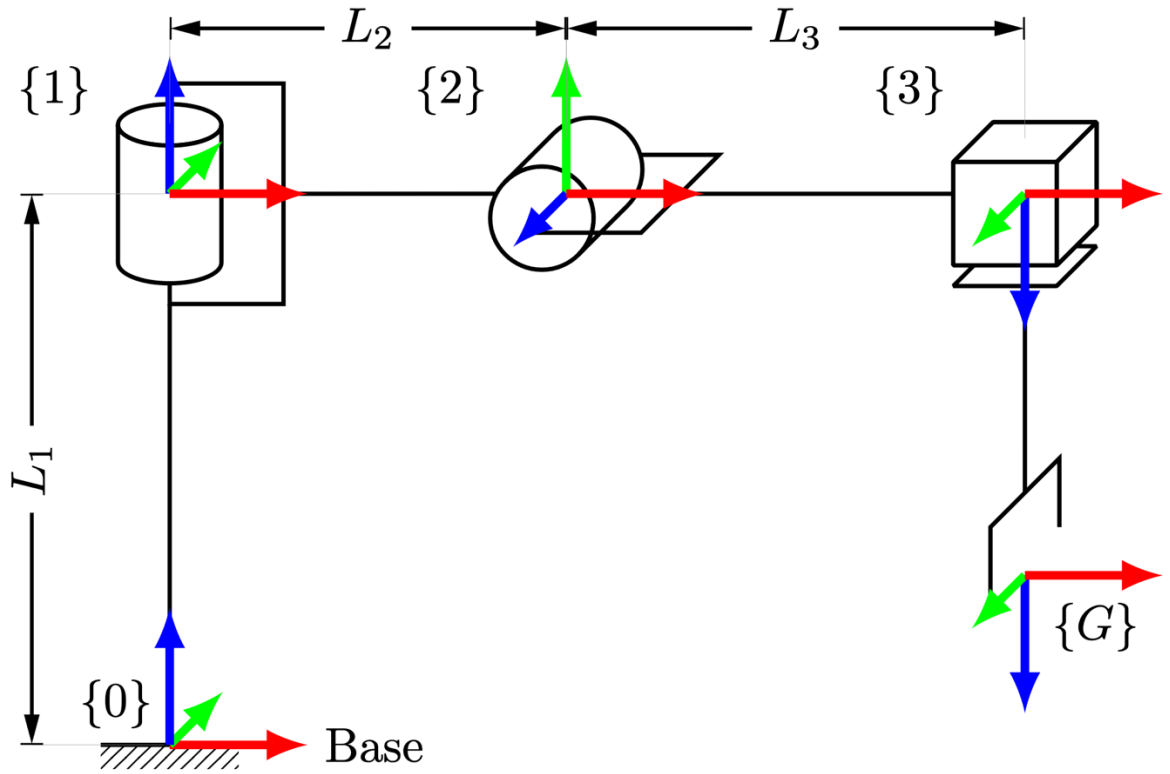


Part II: DH Modeling of a Real Robot

DH Parameters Recap Example (I)



DH Parameters Recap Example (II)

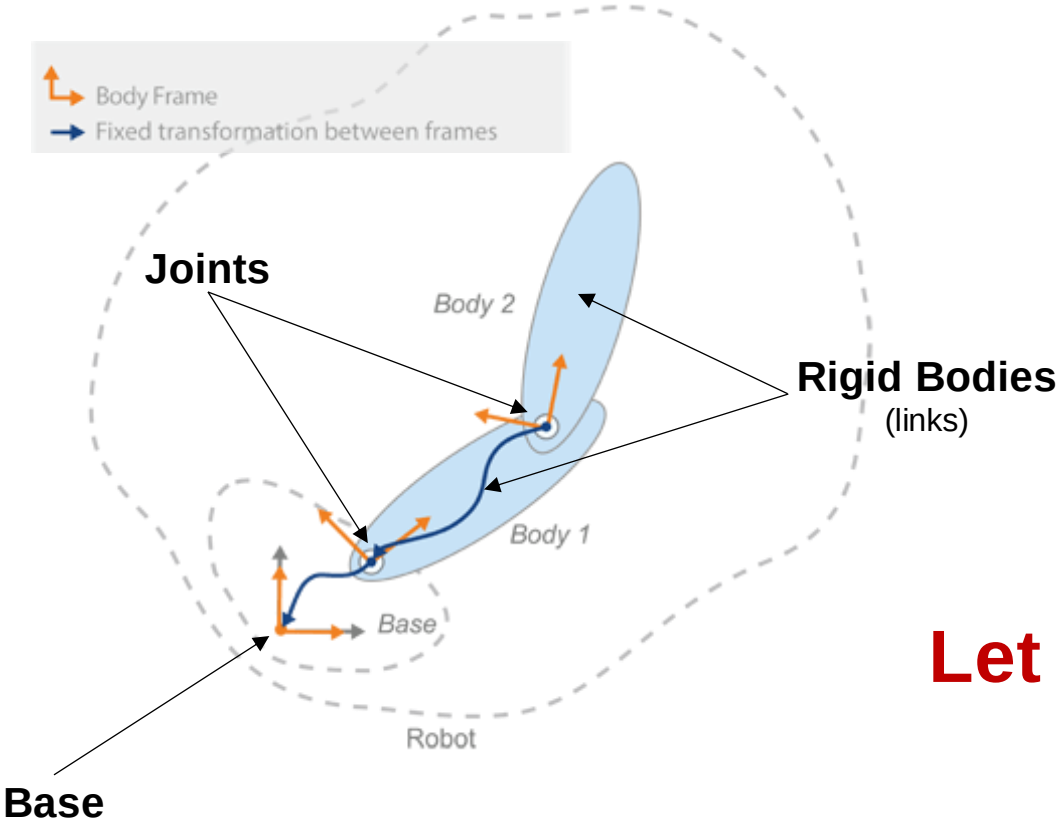


i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	L_1	θ_1
2	90°	L_2	0	θ_2
3	90°	L_3	d_3	0

DH Parameters for MATLAB RigidBodyTrees

MATLAB RigidBodyTree provides a way to **set link transforms based on (Modified) DH Parameters:**

https://se.mathworks.com/help/robotics/ug/build-manipulator-robot-using-kinematic-dh-parameters.html



Let us see it in MATLAB

Agenda for Today

Work individually or in small groups (2-3 people):

Model and simulate a real robot.

1. Draw the robot schematic, assign frames, and derive the Modified DH parameters.
2. Construct a RigidBodyTree simulation from the DH parameters and add visuals with the robot *.stl* files.
3. Use the simulation to calculate the Forward Kinematics.
4. Use simulation to calculate the Numerical Inverse Kinematics.

Go to: *itslearning* → Plans → Lesson 8 → **Point-Giving Assignment 2: Modeling & Simulation of Robots**

You can hand this in before Friday, April 11th 11:59 as PA #2.

Coming soon in *Robotkinematik…*

Aljaz will publish the full Bill of Materials for the robot:

- Some (cheap) **motors**
- **Arduino** board + cables for electronics and control

...together with **code to control the motors from MATLAB**.

On your own, **you can:**

- Buy the parts
- Print the robot link *.stl* files
- Assemble the robot

Then, in a coming *Robotkinematik* lecture, **we can**:

- Connect the Arduino control code with simulation (**digital twin**)
- Use the **FK and IK** in the simulation **to calculate targets** for the robot
- Calculate **linear point to point motions** in MATLAB and use this to move the robot → **Next lecture!**

Thank you for today.

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